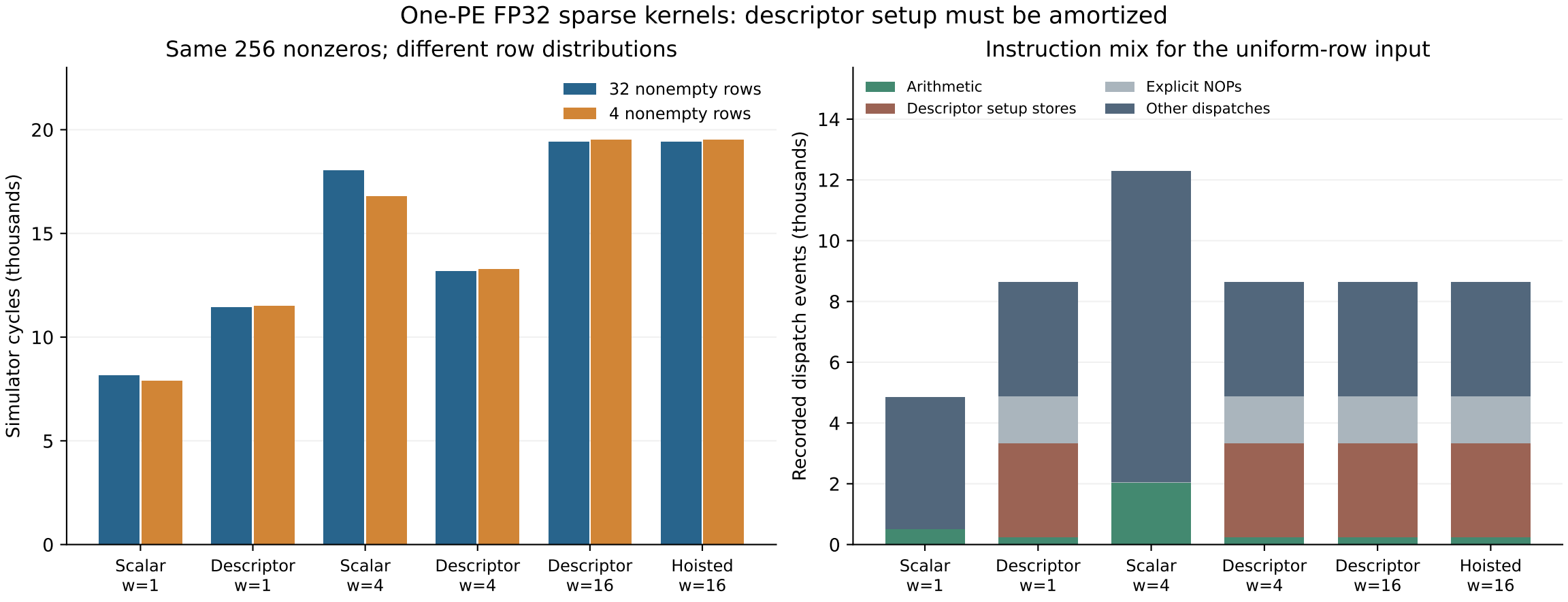
One-PE FP32 sparse kernels: descriptor setup must be amortized
Same 256 nonzeros; different row distributions
Simulator cycles (thousands)
32 nonempty rows
4 nonempty rows
0
5
10
15
20
Scalar w=1
Descriptor w=1
Scalar w=4
Descriptor w=4
Descriptor w=16
Hoisted w=16
Instruction mix for the uniform-row input
Recorded dispatch events (thousands)
Arithmetic
Descriptor setup stores
Explicit NOPs
Other dispatches
0
2
4
6
8
10
12
14
Scalar w=1
Descriptor w=1
Scalar w=4
Descriptor w=4
Descriptor w=16
Hoisted w=16